

Lecture 4: Static Games of Complete Information

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4.1 Static Games of Complete Information

Now we get to talk about actual games.

Example 4.1 (Prisoner's Dilemma) *The setup is*

- 2 players, both criminals who allegedly did a bad crime
- each player can confess (C) or stay silent (S)
- each player is in a different interrogation room
- payoff is based on the actions of both players

$$\begin{array}{l}
 \text{Payoff of Player 1} \quad \underbrace{v_1} \quad \text{Action of player 1} \quad \underbrace{C} \quad , \quad \text{Action of player 2} \quad \underbrace{C} \quad) = -4 = \text{Payoff of Player 2} \quad \underbrace{v_2} \quad \text{Action of player 1} \quad \underbrace{C} \quad , \quad \text{Action of player 2} \quad \underbrace{C} \quad) \\
 v_1(C, S) = -1 = v_2(S, C) \\
 v_1(S, S) = -2 = v_2(S, S) \\
 v_1(S, C) = -5 = v_2(C, S)
 \end{array}$$

We can also represent this as a payoff matrix

		Player 2	
		C	S
Player 1	C	-4, -4	-1, -5
	S	$\underbrace{-5}$, $\underbrace{-1}$	-2, -2
		Player 1 payoff	Player 2 payoff

Table 4.1: The Prisoner's Dilemma as a payoff matrix

We “solve” this game by trying to reason through what rationality looks like here.

For example, if we know Player 2 will choose to confess (c), then we need to figure out what Player 1 will do. This means comparing

$$v_1(c, s) = -4, \quad v_1(s, s) = -5.$$

Then since Player 1 is rational, the best action is for Player 1 to confess since $-4 > -5$.

Now if, Player 2 decides to stay silent (s), then we compare

$$v_1(c, s) = -1, \quad v_1(s, s) = -2.$$

This means that the best action for player 1 is always to confess (c) since $-1 > -2$.

We could also put ourselves in the shoes of Player 2 and see that they would perform the same action.

Example 4.2 (Hanging with Friends¹) Let's say there are two friends who really enjoy spending time with each other, but they have very different hobbies. Player 1 really likes going to art museums (A), player 2 really likes going to a basketball game (B), but they both get more satisfaction being together than being at separate locations. The friends have to decide what they want to do and go to the venue; there is no talking between player 1 and player 2 (think of a time before cell phones).

		Player 2	
		A	B
Player 1	A	2,1	0,0
	B	0,0	1,2

Table 4.2: Hanging with Friends as a payoff matrix

What does rationality look like here?

If player 2 chooses the art museum, A , then player 1 will choose A since $v_1(A, A) = 2 > 0 = v_1(B, A)$. And if player 2 chooses the basketball game B , then player 1 will choose B since $v_1(A, B) = 0 < 1 = v_1(B, B)$. But unlike the last game, there is no definitive “player 1 should always choose XYZ”.

Definition 4.3 (Static) A static game is a game where each player chooses the action simultaneously and independently. No collusion.

Definition 4.4 (Complete information) The following components are common knowledge among all players:

1. all possible actions for all players,
2. all possible outcomes for all players,
3. how each combination of actions affect which outcomes will materialize, and
4. payoffs of all players for all outcomes

Definition 4.5 (Common Knowledge) An event E is common knowledge if

1. everyone knows E ,
2. everyone knows everyone knows E ,
3. everyone knows everyone knows everyone knows E ,
4. etc

Definition 4.6 (Normal-form game) A Normal-Form Game consists of the following features:

1. The set of all players
2. The set of actions for each player
3. the payoff function for each player that gives a payoff value to each combination of the player's chosen action.

Definition 4.7 A pure strategy is a deterministic plan of action fully describing what action a player takes in every possible situation of the game

Example 4.8 (Rock, Paper, Scissors) Let's cast the children's game—Rock, Paper, Scissors—as a normal form game.

		Player 2		
		<i>R</i>	<i>P</i>	<i>S</i>
Player 1	<i>R</i>	0, 0	-1, 1	1, -1
	<i>P</i>	1, -1	0, 0	-1, 1
	<i>S</i>	-1, 1	1, -1	0, 0

Table 4.3: Rock, Paper, Scissors as a payoff matrix

Example 4.9 (Left, Right, Center) Let's play this game: $N = \{\text{Player 1, Player 2}\}$, $S_1 = \{U, M, D\}$, $S_2 = \{L, C, R\}$ with the following payoff matrix.

		Player 2		
		<i>L</i>	<i>C</i>	<i>R</i>
Player 1	<i>U</i>	4, 3	5, 1	6, 2
	<i>M</i>	2, 1	8, 4	3, 6
	<i>D</i>	3, 0	9, 6	2, 8

Table 4.4: Left, Right, Center as a payoff matrix

Let both players be rational. If player 1 plays *U*, player 2 will play *L* since $3 > 2 > 1$. If player 1 plays *M*, player 2 will play *R* since $6 > 4 > 1$. Lastly, if player 1 plays *D*, player 2 will play *R* since $8 > 6 > 0$. This means depending on what player 1 plays, player 2 will either play *L* or *R*, but NEVER *C*.

Why is that?

Definition 4.10 (Strictly dominated strategy) We say that s'_2 is strictly dominated by another strategy s_2 if $v_2(s_1, s_2) > v_2(s_1, s'_2)$ for all action s_1 out of the possible actions player 1 can perform. In other words, whatever player 1 plays, it is always better for player 2 to play s_2 instead of s'_2 .

Remark 4.11 A rational player will never play a strictly dominated strategy. This is because there is always another action they can perform that results in a higher payoff.

Example 4.12 (Left, Right, Center, revisited) We now have the terminology to describe why player 2, as a rational person will never play *C*. This is because *C* is strictly dominated by *R*. Since this is common knowledge, player 1 knows the payoffs and can figure this out. This means the game can be reduced:

		Player 2	
		<i>L</i>	<i>R</i>
Player 1	<i>U</i>	4, 3	6, 2
	<i>M</i>	2, 1	3, 6
	<i>D</i>	3, 0	2, 8

Since player 1 knows that player 2 won't play *C*, then if player 2 plays *L*, player 1 will play *U* since $4 > 3 > 2$. If player 2 plays *R*, then player 1 will play *L* again since $6 > 3 > 2$. This means regardless of what player 2 plays, player 1 will always play *L*. We could also say in this reduced game that both *M* and *D* are strictly dominated by *L*.

But again this is common knowledge that player 1 will do this. This means the game can be reduced again.

		Player 2	
		<i>L</i>	<i>R</i>
Player 1	<i>U</i>	4, 3	6, 2

Since player 1 will always play *U*, then player 2 will always play *L* since $3 > 2$, or *R* is strictly dominated by *L*. This means if our assumptions hold, player 1 will always play *U*, player 2 will always play *L*.

Definition 4.13 (Iterated Elimination of Strictly Dominated Strategy - equilibrium) Any strategy profile that survives this process of iterated elimination of strictly dominated strategies (IESDS) is an iterated-elimination equilibrium or IESDS-equilibrium.

With this redefined notion of “solving a game”, can we solve the hanging with friends game?

Example 4.14 (Hanging with Friends, revisited) If player 2 chooses *A*, then player 1 will choose *A* since $2 > 0$, but if player 2 chooses *B*, then player 1 will choose *B*, since $1 > 0$. But $(2, 0)$ doesn't strictly dominate $(0, 1)$, so we still cannot simplify. Since this game is symmetric, we cannot simplify for player 2 either.

Definition 4.15 (Best Response) If you know or believe that other players will play a certain strategy s_{-i} , then your best response to s_{-i} is the strategy that gives you the maximal payoff.

Remark 4.16 If you knew what the other player was going to play, then of course you can have a best response, but so far we have only looked at games where the players all select a choice simultaneously and independently. (We will discuss dynamic games later). So what good does best response do to us now.

Definition 4.17 (Belief) A belief of player i is a possible strategy profile s_{-i} of the other players.

Definition 4.18 (Best Response, redefined) The best response for a player i to believe s_{-i} is

$$BR_i = \{s_i : v_i(s_i, s_{-i}) \geq v_i(s'_i, s_{-i}) \text{ for all strategies in the set}\}$$